### 12.4.1 Solutions

4. $\frac{\binom{5}{2}\binom{3}{2}}{\binom{8}{2}} = 1.0714286$

5. $4/36 = 1/9$ 0.1111111

6. $4/16 = 1/4$

7. $\frac{6}{36} + \frac{5}{36} - \frac{1}{36}$

8a. $p(\text{large}) = \frac{1}{1000}$; $p(\text{small}) = \frac{6}{1000} - \frac{1}{1000} = \frac{5}{1000}$

8b. $p(\text{large}) = \frac{1}{1000}$; $p(\text{small}) = \frac{3}{1000} - \frac{1}{1000} = \frac{2}{1000}$

8c. $p(\text{large}) = \frac{1}{1000}$; $p(\text{small}) = \frac{0}{1000}$

9. $\frac{\binom{4}{1}\binom{13}{5}}{\binom{52}{5}} = 0$

10. $\frac{\binom{13}{1}\binom{4}{4}\binom{12}{1}\binom{4}{1}}{\binom{52}{5}} = 0.4705882$

11. $1 - \frac{1}{2^8} = 0.0039063$

```
#include: false
ans12a <- 1/6^4
ans12b <- 1 - (6 * 5 * 4 * 3 * 2 / 6^5)
```

12a. $\frac{6}{6^5} = \frac{1}{6^4} = 7.7160494 \times 10^{-4}$

12b. $1 - \frac{6 \cdot 5 \cdot 4 \cdot 3 \cdot 2}{6^5} = 0.9074074$

13a. $\frac{2}{7} =$ `r2/7‘

13b. $\frac{9}{49} = 0.1836735$

13c. $\frac{6}{36} = \frac{1}{6} = 0.1666667$